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**Universal Mobile Telecommunications System (UMTS);**
**LTE;**
**Electromagnetic compatibility (EMC);**
**Table of international requirements for mobile terminals and ancillary equipment**
**(3GPP TR 34.926 version 11.0.1 Release 11)**

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# Foreword

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# Foreword

This Technical Report has been produced by the 3GPP.

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Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 Indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

# Introduction

This Technical Report is a living document and will continuously be upgraded as standards and regulations change. Sections will be added for new regions or nations, which enter the 3GPP and have different requirements. It is the intention to reference global standards but where it is known that none exist then alternative standards mandated by the regional and or national requirements will be quoted.

The sole purpose of the document is as a reference to the current status at time of publication. When new versions are published they will supersede the previous ones.

# 1 Scope

The present document shows in tabular form most of the current regulatory and voluntary requirements by region or nation, and is for information purposes only.

# 2 References

The following documents contain provisions, which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in the same Release as the present document.

[1] IEC 61000-6-1 (1997): "Electromagnetic compatibility (EMC) – Part 6: Generic standards - Section 1: Immunity standard for residential, commercial and light-industrial environments".

[2] IEC 61000-6-3 (1996): "Electromagnetic compatibility (EMC) – Part 6: Generic standards - Section 3: Emission standard for residential, commercial and light-industrial environments.

[3] ISO 7637-1 (1990): "Road vehicles - Electrical disturbance by conduction and coupling - Part 1: Passenger cars and light commercial vehicles with nominal 12 V supply voltage - Electrical transient conduction along supply lines only".

[4] ISO 7637-2 (1990): "Road vehicles - Electrical disturbance by conduction and coupling - Part 2: Commercial vehicles with nominal 24 V supply voltage - Electrical transient conduction along supply lines only".

[5] RECOMMENDATION ITU-R SM.329 " Unwanted emissions in the spurious domain "

[6] IEC CISPR publication 22; 3rd edition (1997-11); "Information technology equipment; Radio disturbance characteristics – Limits and methods of measurement"

[7] IEC CISPR publication 16-1; (1993); Radio disturbance and immunity measuring apparatus"; Am.1 (1997); "Specification for radio disturbance and immunity measuring apparatus and methods"

[8] IEC 61000-3-2; (1995-03); "Electromagnetic compatibility; Part 3 - Limits; section 2 – Limits for harmonic current emissions (equipment input current ≤ 16 A per phase)"; Am.1 (1997-09)

[9] IEC 61000-3-3; (1994-12); "Electromagnetic compatibility; Part 3 - Limits; section 2 – Limitation of voltage fluctuations and flicker in low-voltage supply systems for equipment with rated current ≤ 16 A"

[10] IEC 61000-4-2; "Electromagnetic compatibility (EMC) – Part 4: Testing and measurement techniques – section 2: Electrostatic discharge immunity test – Basic EMC publication"

[11] IEC 61000-4-3; "Electromagnetic compatibility (EMC) – Part 4: Testing and measurement techniques – section 3: Radiated, radio-frequency electromagnetic field immunity test"

[12] IEC 61000-4-4; "Electromagnetic compatibility (EMC) – Part 4: Testing and measurement techniques – section 4: Electrical fast transient/burst immunity test – Basic EMC publication"

[13] IEC 61000-4-5; "Electromagnetic compatibility (EMC) – Part 4: Testing and measurement techniques – section 5: Surge immunity test"

[14] IEC 61000-4-6; "Electromagnetic compatibility (EMC) – Part 4: Testing and measurement techniques – section 6: immunity to conducted disturbances induced by radio frequency fields"

[15] IEC 61000-4-11; "Electromagnetic compatibility (EMC) – Part 4: Testing and measurement techniques – section 11:Voltage dips, short interruptions, and voltage variations immunity test"

[16] EN 300 607 - 1; Digital cellular telecommunications system (Phase 2); Mobile Station (MS) conformance specification; Part 1: Conformance specification (GSM 11.10-1 version 4.24.0)

[17] EN 55022; "Information technology equipment; Radio disturbance characteristics – Limits and methods of measurement"

[18] ARIB STD-T57 ; "ELECTROMAGNETIC COMPATIBILITY (EMC) FOR RADIO EQUIPMENT Edition 2.0 " ARIB STANDARD

[19] RCR STD-27 ; "PERSONAL DIGITAL CELLULAR TELECOMMUNICATION SYSTEM Edition H " ARIB STANDARD

[20] FCC CFR 47 part 15; RADIO FREQUENCY DEVICES, Code of Federal Regulations Title 47, Volume 1, Parts 0 to 19

[21] CWTS TS C404; China Wireless Telecommunication Standard (CWTS); Working Group 1 (WG1); UE and BTS EMC

[22] ETSI EN 301 489; Electromagnetic compatibility and Radio Spectrum Matters /ERM); Electromagnetic Compatibility (EMC) standard for radio equipment and services;
Part 1: "Common technical requirements";
Part 7: "Specific conditions for mobile and portable radio and ancillary equipment of digital cellular radio telecommunications systems (GSM and DCS)";.
Part 8: "Specific conditions for GSM base stations";
Part 18: "Specific conditions for Terrestrial Trunked Radio (TETRA)"
Part 23: " Specific conditions for 3rd Generation Partnership project (UMTS) Base station radio and ancillary equipment
Part 24: " Specific conditions for 3rd Generation Partnership project (UMTS) for Mobile and potable (UE) radio and ancillary equipment

# 3 Definitions and abbreviations

## 3.1 Definitions

**Ancillary equipment:** Equipment (apparatus), used in connection with a user equipment (UE) is considered as an ancillary equipment (apparatus) if:

- the equipment is intended for use in conjunction with a UE to provide additional operational and/or control features to the UE, (e.g. to extend control to another position or location); and
- the equipment cannot be used on a stand alone basis to provide user functions independently of a UE; and
- the UE to which it is connected, is capable of providing some intended operation such as transmitting and/or receiving without the ancillary equipment (i.e. it is not a sub-unit of the main equipment essential to the main equipment basic functions).

**Idle mode:** Idle mode is the state of User Equipment (UE) when switched on but with no Radio Resource Control (RRC) connection.

**Port**: particular interface, of the specified equipment (apparatus), with the electromagnetic environment. For example, any connection point on an equipment intended for connection of cables to or from that equipment is considered as a port (see figure 1).

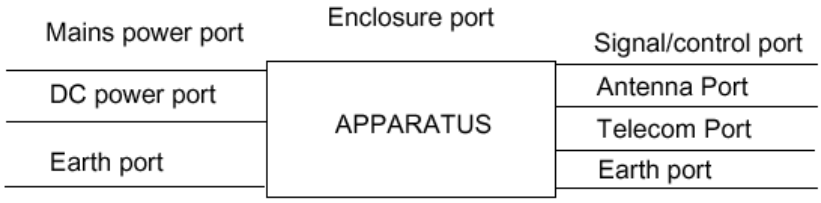


**Figure 1: Examples of ports**

**Spurious emission from ITU-R SM 329:** Emission on a frequency, or frequencies, which are outside the necessary bandwidth and the level of which may be reduced without affecting the corresponding transmission of information. Spurious emissions include harmonic emissions, parasitic emissions, intermodulation products and frequency conversion products but exclude out-of-band emissions.

NOTE For multi-channel or multi-carrier transmitters/transponders, where several carriers may be transmitted simultaneously from a final output amplifier or an active antenna, the centre frequency of the emission is taken to be the centre of the –3dB bandwidth of the transmitter or transponder.

**Telecommunication port:** Ports which are intended to be connected to telecommunication networks (e.g. public switched telecommunication networks, integrated services digital networks), local area networks (e.g. Ethernet, Token Ring) and similar networks (see CISPR 22).

**Transient phenomena:** Pertaining to or designating a phenomena or a quantity which varies between two consecutive steady states during a time interval short compared with the time-scale of interest (IEC 60050-161 ).

# 4 Table of International EMC Requirements for Mobile terminals and ancillary equipment

| SPEC ITEM AREA | APPLIC. | China | | 3GPP SPECS | | JAPAN | | EUROPE | | USA | | KOREA | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | REQMENT | REF. DOC | REQMENT | REF. DOC | REQMENT | REF. DOC | REQMENT | REF. DOC | REQMENT | REF. DOC | REQMENT | REF. DOC |
| **RADIATED EMISSIONS** | | | | | | | | | | | | | |
| Limits on Radiated Emission | Vehicular, Portable, Ancillary, | TS C404 | TS 34.124 TS 25.113 | | TS 34.124 | **classA(Q-peak)** 30MHz-230 MHz : 40dBuV/m 230MHz-1GHz : 47 dBuV/m **classB(Q-peak)** 30 MHz -230 MHz : 30 dBuV/m 230MHz-1GHz : 37 dBuV/m | ARIB T-57 2.2 reference from CISPR22 *(measurement is under receiving condition) | | EN 300 607-1 Version 6 release 1997 | | | **class A (Q-peak)** 30 MHz-230 MHz : 40dBuV/m 230MHz-1GHz : 47 dBuV/m **classB (Q-peak)** 30 MHz -230 MHz : 30 dBuV/m 230MHz-1GHz : 37 dBuV/m | CISPR22 |
| | | | | | | | | | | | | | |
| Transmit OFF power (idle m ode) | | | TS 34.124 TS 25.113 | | TS 34.124 | -60 dBm for PDC terminal | RCR 27 (Standard for PDC) | | EN 300 607-1 Version 6 release 1997 | | | | |
| | | | | | | | | | | | | | |
| Limits on Radiated Emissions | Ancillary AC eqpt only | | | | TS 34.124 | | CISPR 22 | | EN 55022 | Not Req for Part 24 devices. 500uV/m >960 MHz or CISPR 22 | CFR 47 Part 15.109(a),(e) needs further investigation | | |
| | | | | | | | | | | | | | |
| **CONDUCTED EMISSIONS** | | | | | | | | | | | | | |
| A. Limits on Conducted Emission | Ancillary equipment | | TS 34.124 TS 25.113 | | TS 34.124 | | | | | Not Reqd for Part 24 devices. | | | |
| DC Power in/out | Vehicular, Portable, Ancillary | | TS 34.124 TS 25.113 | | TS 34.124 | 0.15 - 0.5 MHz : Q-Peak 79dBuV Average 66dBuV 0.5 - 30 MHz : Q-Peak 73dBuV Average 60dBuV | ARIB T-57 2.3 reference from CISPR 16-1 / 22 | | CISPR 16-1, EN55022 | | | 0.15 - 0.5 MHz : Q-Peak 79dBuV Average 66dBuV 0.5 - 30 MHz : Q-Peak 73dBuV Average 60dBuV | CISPR22 |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AC Mains | Portable, Ancillary | | TS 34.124 TS 25.113 | | TS 34.124 | Class A 0.15 - 0.5 MHz : Q-Peak 79dBuV Average 66dBuV 0.5 - 30 MHz : Q-Peak 73dBuV Average 60dBuV Class B 0.15 - 0.5 MHz : Q-Peak 66-56 dBuV / Average 56-46 dBuV 0.5 - 5 MHz : Q-Peak 56 dBuV Average 46dBuV 5 -30 MHz : Q-Peak 60 dBuV Average 50dBuV | ARIB T-57 2.4 reference from CISPR 16-1 / 22 | | CISPR 22, EN55022 | 250 uV <30 MHz or CISPR 22 | CFR part 15.107(a), (e) | Class A 0.15 - 0.5 MHz : Q-Peak 79dBuV Average 66dBuV 0.5 - 30 MHz : Q-Peak 73dBuV Average 60dBuV Class B 0.15 - 0.5 MHz : Q-Peak 66-56 dBuV / Average 56-46 dBuV 0.5 - 5 MHz : Q-Peak 56 dBuV Average 46dBuV 5 -30 MHz : Q-Peak 60 dBuV Average 50dBuV | CISPR22 |
| Harmonic Current Emissions, AC Mains | Portable, Ancillary | | TS 34.124 TS 25.113 | | TS 34.124 | Not Applicable | | | EN 61000-3-2 | | | Not Applicable | |
| Voltage Fluctuations/Flicker | Portable, Ancillary | | TS 34.124 TS 25.113 | | TS 34.124 | Not Applicable | | | EN 61000-3-3 | | | Not Applicable | |
| | | | | | | | | | | | | | |
| | | | | | | | | | | | | | |
| Immunity to RF EM Fields, 80-1000 MHz | Vehicular, Portable, Ancillary eqpt, | | RF EM Fields, 80-2000 MHz | | TS 34.124 | 3 V/m | ARIB T-57 3.6 reference from JIS 1000-4-3 (IEC 61000-4-3) | 3 V/m | EN 61000-4-3 | | | 3 V/m (80MHz~1GHz: No modulation) | IEC 61000-4-3 |
| | | | | | | | | | | | | | |
| | | | | | | | | | | | | | |
| AC Mains - Voltage Dips and interruption | Portable, Ancillary eqpt, | | TS 34.124 TS 25.113 | | TS 34.124 | Not applicable | | | EN 61000-4-11 | | | Under Consideration | IEC 61000-4-11 |
| AC Mains - Surges, Common Mode and Differental mode | Portable, Ancillary eqpt, | | TS 34.124 TS 25.113 | | TS 34.124 | Not applicable | | | EN 61000-4-5 | | | 1 kV | IEC 61000-4-5 |
| DC Mains-Surges | Vehicular, Portable, Ancillary eqpt, | | TS 34.124 TS 25.113 | | TS 34.124 | Not applicable | | | | | | Under Consideration | IEC 61000-4-5 |
| Signal ports and Comm-unication ports-Surges | Vehicular, Portable, Ancillary eqpt, | | TS 34.124 TS 25.113 | | | Not applicable | | | | | | | |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DC Mains-Surges | Vehicular, | | TS 34.124 TS 25.113 | | TS 34.124 | 50 V / -50 V and -5 V / -2.5 V / 0 V | ARIB T-57 3.10 reference from ISO 7637-1/2 | | ISO 7637-1/2 | | | Under Consideration | |
| Fast Transients - AC/DC Power | Portable, Ancillary eqpt, | | TS 34.124 TS 25.113 | | TS 34.124 | Not applicable | | 2/1 Kv | EN 61000-4-4 | | | 1kV | IEC 61000-4-4 |
| Fast Transients - Signal/control ports | Portable, Ancillary eqpt, | | TS 34.124 TS 25.113 | | TS 34.124 | Not applicable | | 0.5 Kv | EN 61000-4-4 | | | 0.5kV | IEC 61000-4-4 |
| | | | | | | | | | | | | | |
| RF Conducted 0.15-80MHz | Vehicular, Portable, Ancillary eqpt, | | TS 34.124 TS 25.113 | | TS 34.124 | 3 Vrms | ARIB T-57 3.7 v. 2.0 reference from IEC 61000-4-6 (only Vehicular) | 3 V rms | EN 61000-4-6 | | | Under Consideration | IEC 61000-4-6 |
| | | | | | | | | | | | | | |
| ELECTRO-STATIC DISCHARGE | Vehicular, Portable, Ancillary eqpt, | | TS 34.124 TS 25.113 | | TS 34.124 | ±8 kV (Air) / ± 4 kV (Contact) | ARIB T-57 3.4 v.2.0 reference from IEC 61000-4-2 | ±8 kV (Air) / ± 4 kV (Contact) | EN 61000-4-2 | | | ±8 kV (Air) / ± 4 kV (Contact) | IEC 61000-4-2 |
| | | | | | | | | | | | | | |
| | | | TS 34.124 TS 25.113 | | TS 34.124 | 1 A/m (60 Hz or 50 Hz) | ARIB T-57 3.8 v.2.0 reference from IEC 61000-4-8 | | | | | | |
| | | | | | | | | | | | | | |

# Annex A (informative): Change history

| Change history | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Time | Doc | CR | R | Ph | Title | Cat | Curr | New | Work Item |
| RP-36 | | | | | Creation of v7.0.0 based on version 6.0.0 | | 6.0.0 | 7.0.0 | |
| RP-42 | | | | | Upgraded unchanged from Rel-7 | | 7.0.0 | 8.0.0 | |
| SP-46 | | | | | Upgraded unchanged from Rel-8 | | 8.0.0 | 9.0.0 | |
| SP-51 | | | | | Upgraded unchanged from Rel-9 | | 9.0.0 | 10.0.0 | |
| SP-57 | - | - | - | - | Update to Rel-11 version (MCC) | - | 10.0.0 | 11.0.0 | - |
| 2012-11 | | | | | Correction of typo on cover page | | 11.0.0 | 11.0.1 | - |

# History

| Document history | | |
|---|---|---|
| V11.0.1 | November 2012 | Publication |
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